

Group Project 3

Merton Jump-Diffusion: Pricing and Harvesting the Equity Jump Risk Premium

Computational Finance Course

11 Students — Week 4: Endterm — Week 6: Final

1 Overview and Motivation

Equity markets do not move continuously. On **27 October 1997** the Dow Jones fell 7% in one day. In **March 2020** the S&P 500 dropped 34% in 23 trading days. These discontinuities — jumps — are a defining feature of equity returns that Black–Scholes cannot capture. Investors who bear jump risk demand a *jump risk premium*, and this premium can be harvested systematically.

The **Merton (1976) jump-diffusion model** is the first and simplest model to capture this behaviour: it adds a compound Poisson jump process to geometric Brownian motion. Despite its simplicity, it remains widely used in credit, equity, and FX derivatives, and is the conceptual foundation for all more complex Lévy-based models.

Career Relevance

Understanding jump models and the jump risk premium is essential for: equity derivatives traders, credit quants (Merton credit model uses jumps), risk managers computing tail VaR, and quant researchers developing systematic vol strategies. Pre-earnings vol selling — the strategy you will backtest — is actively traded by many quant hedge funds.

2 Asset and Models

- **Asset:** NASDAQ-100 (QQQ) or Russell 2000 (IWM) European options. These indices have higher jump frequencies than SPX.
- **Benchmark:** Black–Scholes (Geometric Brownian Motion).
- **Main model:** Merton (1976) jump-diffusion.
- **Extension:** Kou (2002) double-exponential jump model (compare fits).

3 Mathematical Framework

Under $\mathbb{Q}$, the Merton model is:

$$\frac{dS_t}{S_t} = (r - \lambda \bar{\mu}) dt + \sigma dW_t + (e^{J_k} - 1) dN_t, \quad (1)$$

where $N_t \sim \text{Poisson}(\lambda t)$, $J_k \sim \mathcal{N}(\mu_J, \sigma_J^2)$ i.i.d., and $\bar{\mu} = e^{\mu_J + \sigma_J^2/2} - 1$.

Exact analytic formula (Merton 1976): the call price is a Poisson-weighted sum of Black–Scholes prices:

$$C^{\text{Merton}}(S, K, T) = \sum_{n=0}^{\infty} \frac{e^{-\lambda'T} (\lambda'T)^n}{n!} C^{BS}(S, K, T, r_n, \sigma_n), \quad (2)$$

where $\lambda' = \lambda(1 + \bar{\mu})$, $r_n = r - \lambda\bar{\mu} + n\mu_J/T + n\sigma_J^2/(2T)$, $\sigma_n^2 = \sigma^2 + n\sigma_J^2/T$.

Parameters:

Symbol	Meaning	Typical equity range
σ	Diffusion volatility	[0.10, 0.40]
λ	Jump intensity (jumps/year)	[0.5, 8]
μ_J	Mean log-jump size	[−0.15, 0.05]
σ_J	Jump size std	[0.05, 0.30]

4 Team Structure

W	Role	Deliverable
W1	Research Lead (1–2)	Literature survey, jump literature frontier
W2	Theory Lead (1–2)	Merton derivation, Kou model comparison
W3	MC Engineer (1–2)	Monte Carlo with variance reduction
W4	Fourier Engineer (1)	COS engine + exact formula validation
W5	Data Scientist (1–2)	QQQ data, jump detection
W6	Calibration Eng (1–2)	Merton calibration, Kou comparison
W7	Vol Analyst (1)	Smile limitation analysis
W8	Risk Lead (1–2)	Jump hedging, earnings backtest

5 Week 4 — Endterm Requirements

5.1 W1: Literature Review

Required:

- Merton (1976): original jump-diffusion paper.
- Kou (2002): double-exponential jump distribution.
- Carr–Wu (2003): finite activity vs. infinite activity Lévy processes.
- Eraker, Johannes, Polson (2003): MCMC estimation of jump parameters from returns data.
- Two recent papers (2020–2024): machine learning jump detection, Hawkes self-exciting processes, or deep hedging with jumps.

Frontier: (1) Self-exciting Hawkes jumps (crashes cluster in time); (2) rough volatility + jumps; (3) neural jump-diffusion estimation from options.

5.2 W2: Mathematical Derivations

1. Derive equation (1) from the Lévy–Itô decomposition. Show the stock price discounted by e^{-rt} is a $\mathbb{Q}$ -martingale.
2. Derive the Merton characteristic function: $\phi^{\text{Merton}}(u) = \exp\left(iu(r - \frac{1}{2}\sigma^2 - \lambda\bar{\mu})T - \frac{1}{2}\sigma^2u^2T + \lambda T(e^{i\bar{\mu}u} - 1 - \frac{1}{2}\sigma^2u^2)\right)$.
3. Prove the Merton pricing formula (Poisson-weighted BS sum).
4. **Market incompleteness:** prove rigorously that when jumps are present, there is no unique equivalent martingale measure — the market is incomplete. Explain the practical implication for hedging.
5. **Kou model:** state the double-exponential distribution for jump sizes; write its characteristic function; explain why it has a closed-form formula for barrier and lookback options (Kou–Wang 2003 result).

5.3 W3: Monte Carlo Engine

Implement `merton_mc.py`:

1. Simulate GBM paths + compound Poisson jumps.
2. **Thinning algorithm:** draw Poisson number of jumps on each path; add lognormal jump sizes at uniform jump times.
3. **Variance reduction (all three required):** (a) antithetic variates, (b) control variate using the exact Merton formula, (c) moment matching (adjust paths so sample mean = e^{rT}).
4. Produce a variance reduction table: show standard error reduction vs. plain MC for each method.
5. **Convergence study:** RMSE vs. paths N and vs. time steps M .
6. **Comparison:** price a call with (a) GBM only, (b) Merton, and (c) Merton with large negative jumps ($\mu_J = -20\%$). Show implied vol smile for each case.

5.4 W4: COS Pricing Engine

Implement `merton_cos.py`:

1. COS method with Merton characteristic function.
2. **Triple validation:** COS result must agree with (a) exact Merton formula (to machine precision, $< 10^{-8}$ error) and (b) MC (within Monte Carlo error bands).
3. Extend to Kou model: implement the double-exponential characteristic function and price options; compare Merton vs. Kou smile shapes.
4. Digital option pricing using COS.

5.5 W5: Data Collection and Jump Detection

1. Collect QQQ option chains: at least 6 maturities, 15+ strikes each.
2. Download 5-minute QQQ returns for the past 3 years.
3. **Realised variance decomposition:** compute

$$RV_t = \sum_i r_{t,i}^2, \quad BPV_t = \mu_1^{-2} \sum_i |r_{t,i}| |r_{t,i+1}|,$$

where $\mu_1 = \sqrt{2/\pi}$. The ratio test $Z_t = \sqrt{n}(RV_t/BPV_t - 1)/\hat{\omega}$ detects significant jumps (Barndorff-Nielsen–Shephard test).

4. Report: estimated jump frequency, average jump size, jump size distribution (compare to lognormal assumption).
5. Identify major jump dates (earnings surprises, macro events); annotate on a time series plot.

6 Week 6 — Final Requirements

6.1 W6: Calibration

1. Calibrate Merton $(\sigma, \lambda, \mu_J, \sigma_J)$ to QQQ implied vol surface (minimise vol-space RMSE).
2. **Identification problem:** σ and λ are not jointly identified from a single option price — run a study showing multiple (σ, λ) pairs that produce the same ATM vol; explain why this is a problem and how regularisation helps.
3. Calibrate Kou model on the same data; compare RMSE (Merton vs. Kou).
4. **Empirical vs. risk-neutral jump parameters:** compare calibrated λ (risk-neutral) to BNS-estimated λ^P (physical). The difference $\lambda - \lambda^P$ measures the **jump risk premium**.

6.2 W7: Implied Volatility Surface and Model Limitations

1. Plot Merton vol surface vs. QQQ market surface; residual surface.
2. **Smile decay:** prove analytically that the Merton smile decays as $O(1/\sqrt{T})$ for large T but the market smile decays more slowly. This is the fundamental limitation of the model.
3. Compare smile shape at short maturities (1M): where does Merton fit well vs. where does it fail?
4. Discuss which richer model addresses each failure (hint: Heston + jumps = Bates; infinite activity Lévy for short-maturity behaviour).

6.3 W8: Hedging and Jump Risk P&L

Part A — Minimum Variance Hedge:

1. Under Merton, the minimum variance hedge ratio is not just Delta; it requires a correction for jump risk. Derive and implement:

$$\Delta^{\text{MV}} = \frac{\partial C}{\partial S} + \frac{\lambda \mathbb{E}[(e^J - 1)\Delta C^{\text{jump}}]}{\sigma^2 S},$$

where ΔC^{jump} is the expected option price change due to a jump.

2. Compare P&L distributions under (a) BS Delta, (b) Merton Delta, (c) Merton minimum-variance hedge.

Part B — Tail Risk:

1. Show that the hedging error distribution is **non-Gaussian** under jump-diffusion. Compute kurtosis; compare to the BS case.
2. Plot the 5% left tail of P&L distribution across all three hedges.

6.4 Alpha: Pre-Earnings Implied Volatility Collapse

Alpha Section

Implied volatility systematically inflates before earnings announcements and collapses afterwards — regardless of the earnings outcome. This is one of the most robust short-term alpha signals in equity derivatives.

Strategy: Short Straddle Into Earnings, Close Post-Announcement

1. **Data:** download earnings announcement dates for NASDAQ-100 constituents (at least 50 stocks, 3 years). Use `yfinance` calendar or `earnings-calendar` library.
2. **Signal:** on day $t = -1$ before earnings, compute the implied vol “earnings premium”: $EP_t = \sigma_{\text{nearest expiry}}^{\text{IV}} - \sigma_{\text{next expiry}}^{\text{IV}}$. Sell a straddle only if $EP_t > 5\%$.
3. **Trade:** sell ATM straddle at close of $t = -1$; close position at open of $t = +1$ (post-announcement).
4. **Merton model use:** calibrate Merton with an announcement-specific jump $\lambda_{\text{earnings}}$ separately from the background jump intensity; show the model captures the earnings premium accurately.
5. **Backtest:** compute per-trade P&L, aggregate Sharpe ratio, win rate, average hold P&L as fraction of premium collected.
6. **Jump beta:** regress strategy P&L on a “jump beta” factor (realized minus expected jumps); show the strategy has negative jump beta (it loses when actual earnings jumps exceed model expectation).
7. **Factor regression:** $R_t^{\text{strategy}} = \alpha + \beta_1 R_t^{\text{QQQ}} + \beta_2 \Delta \text{VIX}_t + \beta_3 \text{JumpFactor}_t + \varepsilon_t$.

Output: `earnings_vol_backtest.py`; per-stock P&L breakdown; Sharpe table; factor regression; comparison of Merton-model vs. BS-model premium estimates.

7 Backtesting Standards

- Use only options data available at time of trade (no look-ahead).
- Include bid–ask spread: use half-spread from market data.
- Transaction cost: assume 0.15 vol point per trade (options are less liquid around earnings).
- Report per-strategy-event P&L; never aggregate across positions held simultaneously (isolate the earnings alpha).
- Stress events: document large losses (e.g., when earnings surprise creates a move $> 3\sigma$); show that stop-losses help.

8 Grading Rubric

8.1 Week 4 Endterm (50%)

Component	Weight	Criterion
Literature & frontier	15%	Lévy processes covered
Derivations	20%	Incompleteness proof, Kou CF
MC engine	25%	Variance reduction table
COS engine	20%	Triple validation
Data & jump detection	20%	BNS test, jump parameter estimation

8.2 Week 6 Final (50%)

Component	Weight	Criterion
Calibration	20%	Identification, jump risk premium
Vol surface	15%	Smile decay analysis
Hedging & P&L	20%	MV hedge, tail risk
Alpha / backtest	25%	Earnings strategy, factor beta
Paper quality	20%	Research paper standards

9 Key References

- Merton, R.C. (1976). Option Pricing When Underlying Stock Returns are Discontinuous. *Journal of Financial Economics*, 3, 125–144.
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